EXAMINATION QUESTION PAPER - CONFIDENTIAL LEAK

Subject: Computer Science & Artificial Intelligence

Q1. To guarantee finding the optimal path in A* tree search, what property must the heuristic h

A. Consistency: h(n) <= c(n, a, n') + h(n')

B. Admissibility: h(n) must never overestimate the true cost to reach the goal

C. Monotonicity: h(n) must monotonically increase along all search paths

D. Strict positivity: h(n) must always be greater than zero for all nodes

Q2. For an unweighted graph, which traversal algorithm always finds the shortest path?

A. Depth-Limited Search (DLS)

B. Breadth-First Search (BFS)

C. Depth-First Search (DFS)

D. Greedy Best-First Search

Q3. What is the primary rationale for scaling the query-key product by 1/sqrt(d_k) in Transform

A. To decrease the computational matrix complexity from O(N^2) to O(N)

B. To prevent extremely large dot products from pushing softmax into regions with vanishingly s

C. To ensure the resulting attention weights sum up to exactly one across sequence length

D. To normalize the sentence embeddings across batch dimensions during training